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FACULTY OF MECHANICAL ENGINEERING AND ROBOTICS

Service Robots – Project

SCANNER ROBOT

Pojasek Wojciech

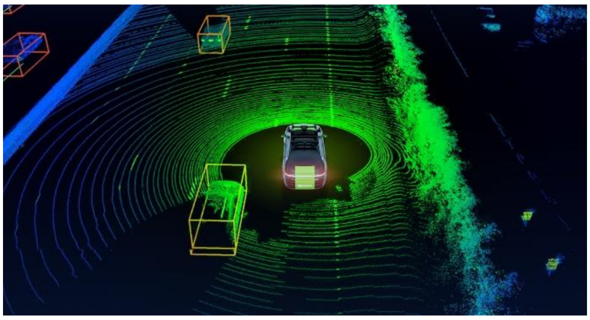
Index Number: 419773

I. State of Art

Nowadays, automation and robotics are rapidly developing in the field of industrial design and quality control. One of the key directions of progress is the use of robotic manipulators for 3D object scanning and reverse engineering. These systems aim to automate the process of converting physical objects into accurate digital 3D models, reducing human error and increasing precision.

Traditional 3D scanning setups often require manual operation or static scanner positioning, which limits flexibility and accuracy for complex shapes. Integrating LIDAR sensors with a robotic arm allows the robot to autonomously analyze the geometry of an object from multiple angles, ensuring high-resolution scanning and efficient data fusion into a 3D CAD model in STP format so all the softwares like Inventor, SolidWorks etc. are able to open it.

LIDAR SENSOR



1. Existing Solutions

At present, there are several examples of robots capable of performing scanning or inspection tasks, but very few that achieve fully autonomous reverse engineering. As an illustrative case, the KALMAN robotic rover can be mentioned — a Mars-rover-like platform capable of navigating and performing environmental mapping using LiDAR sensors. Although it is not directly focused on 3D model reconstruction, it represents the integration of mobile robotics with spatial scanning technology, which can inspire further development of such systems.

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2. Future Improvements of a Concept

Future improvements could include the combination of the robotic scanning arm with a mobile platform, allowing it to autonomously move around large objects. Another possible direction is to implement AI-based algorithms for automatic feature recognition and CAD model reconstruction, transforming raw point cloud data directly into parametric solid models. Integration with cloud computing and digital twin systems could further enhance processing speed, data management, and collaborative design.

3. Advantages of this concept

High scanning precision:
The use of a LIDAR sensor ensures accurate distance measurement and reliable 3D reconstruction, even for complex geometries.

Automation of reverse engineering:
The robotic arm performs scanning without manual intervention, reducing human error and improving repeatability.

Flexible object handling:
Thanks to multiple degrees of freedom, the arm can reach various object angles and surfaces that stationary scanners cannot.

Scalability and adaptability:
The concept can be adapted for different object sizes, materials, and industrial environments by modifying the arm's dimensions or sensor type.

Integration with CAD systems:
The system outputs directly to standard engineering formats (e.g., *.STP*), enabling seamless use in design, prototyping, and quality control workflows.

Potential for AI enhancement:
Data collected by the LIDAR can be processed using machine learning algorithms for automatic feature recognition and surface reconstruction.

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4. Disadvantages of this concept

Complex calibration:
Achieving high-accuracy scanning requires precise calibration between the robot's kinematics and the LIDAR sensor frame.

High computational demand:
Processing large point clouds and generating 3D models in real time requires powerful hardware or cloud computing resources.

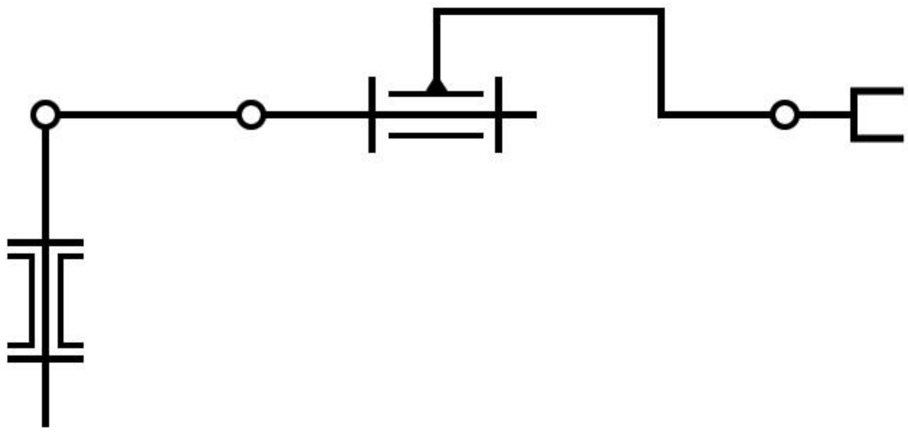
Cost of components:
LIDAR sensors, high-torque servomotors, and precision encoders significantly increase the total system cost compared to standard manipulators.

Limited speed:
Due to the detailed scanning process and data acquisition rate, the operation time can be relatively long for large or complex objects.

Environmental sensitivity:
LIDAR performance can be affected by ambient lighting, surface reflectivity, or dust, which may reduce data quality.

Data fusion complexity:
Combining multiple scans into a single coherent 3D model requires advanced algorithms for alignment and noise reduction.

II. Kinematic Scheme



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We can determine a number of robot's degrees of freedom with the use of formula:

$$M = 6n - \sum_{i=1}^{j}(6 - f_i)$$

n-number of links

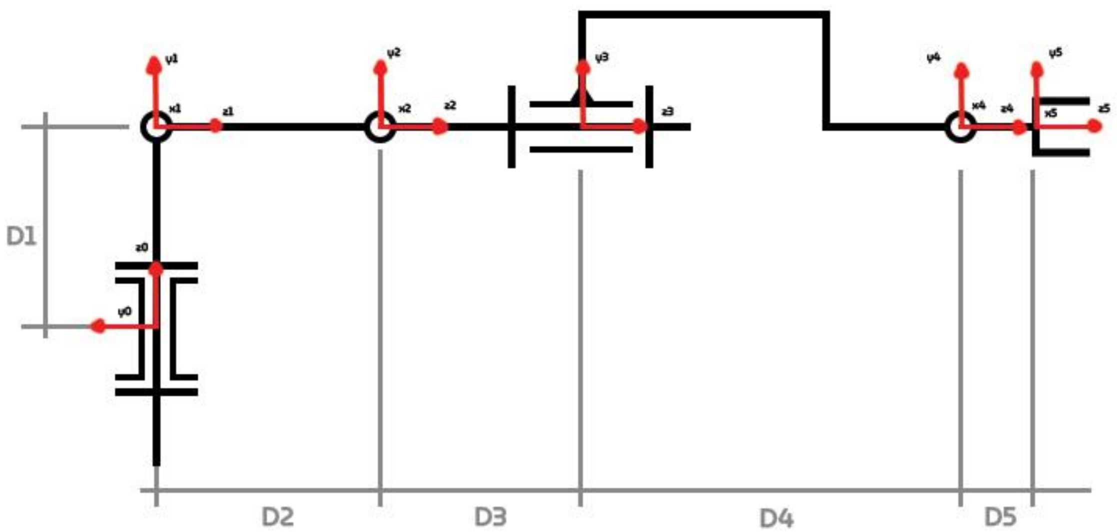
j-number of joints

In case of my robot will look as follows:

M = 6*5 – 5(6-1) = 5

III. Denavit-Hartenberg system notation

This convention is used for selecting frames of reference, which gives a chance to write the kinematic equations of specific manipulator.



	θ	**d**	**a**	**α**
1	θ_1_var	d_1	0	90
2	0	0	0	α_2_var
3	0	d_2	0	α_3_var
4	0	d_3_var	0	0
5	0	d_4	0	α_5_var
6	0	d_5	0	0

Notation:

θ_i – rotation about z-axis
d_i – translation along z-axis
a_i – rotation about x-axis
α_i – translation along x-axis

$d_1 = 0$
$d_2 = 560$
$d_3 = 440$
$d_4 = 390$
$d_5 = 520$

IV. Forward and inverse kinematics

a) Forward

Global transformation matrix equation:

$$T_6 = A_1 A_2 A_3 A_4 A_5 A_6 = \begin{matrix} o_{x1} & o_{x2} & o_{x3} & p_x \\ o_{y1} & o_{y2} & o_{y3} & p_y \\ o_{z1} & o_{z2} & o_{z3} & p_z \\ 0 & 0 & 0 & 1 \end{matrix}$$

Where:

p- position of end effector

o-orientation of end effector

General equation for end effector position matrix:

$$A_i = Rot(z,\theta_i) \cdot Trans(0,0,d_i) \cdot Trans(a_i,0,0) \cdot Rot(x,\alpha_i)$$

$$A_i = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i)\cos(\alpha_i) & \sin(\theta_i)\sin(\alpha_i) & a_i\cos(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i)\sin(\alpha_i) & -\cos(\theta_i)\sin(\alpha_i) & a_i\sin(\theta_i) \\ 0 & \sin(\alpha_i) & \cos(\alpha_i) & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

In my system:

$$A_1 = Rot_{z,\theta_1} \cdot Trans_{z,d_1} \cdot Rot_{x,90}$$

$$A_1 = \begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & 0 \\ \sin(\theta_1) & \cos(\theta_1) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(90) & -\sin(90) & 0 \\ 0 & \sin(90) & \cos(90) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} \cos(\theta_1) & 0 & \sin(\theta_1) & 0 \\ \sin(\theta_1) & 0 & -\cos(\theta_1) & 0 \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2 = Rot_{x,\alpha_2} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\alpha_2) & -\sin(\alpha_2) & 0 \\ 0 & \sin(\alpha_2) & \cos(\alpha_2) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3 = Trans_{z,d_2} \cdot Rot_{x,\alpha_3}$$

$$A_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\alpha_3) & -\sin(\alpha_3) & 0 \\ 0 & \sin(\alpha_3) & \cos(\alpha_3) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\alpha_3) & -\sin(\alpha_3) & 0 \\ 0 & \sin(\alpha_3) & \cos(\alpha_3) & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_4 = Trans_{z,d_3} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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$$A_5 = Trans_{z,d_4} \cdot Rot_{x,\alpha_5}$$

$$A_5 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\alpha_5) & -\sin(\alpha_5) & 0 \\ 0 & \sin(\alpha_5) & \cos(\alpha_5) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\alpha_5) & -\sin(\alpha_5) & 0 \\ 0 & \sin(\alpha_5) & \cos(\alpha_5) & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_6 = Trans_{z,d_5} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T = A_1 A_2 A_3 A_4 A_5 A_6$$

$$T = \begin{bmatrix} \cos(\theta_1) & \sin(\theta_1)\sin(\alpha_2+\alpha_3+\alpha_5) & \cos(\alpha_2+\alpha_3+\alpha_5)\sin(\theta_1) & \sin(\theta_1)(d_3\cos(\alpha_2+\alpha_3)+d_4\cos(\alpha_2+\alpha_3)+d_2\cos(\alpha_2)+d_5\cos(\alpha_2+\alpha_3+\alpha_5)) \\ \sin(\theta_1) & -\cos(\theta_1)\sin(\alpha_2+\alpha_3+\alpha_5) & -\cos(\theta_1)\cos(\alpha_2+\alpha_3+\alpha_5) & -\cos(\theta_1)(d_3\cos(\alpha_2+\alpha_3)+d_4\cos(\alpha_2+\alpha_3)+d_2\cos(\alpha_2)+d_5\cos(\alpha_2+\alpha_3+\alpha_5)) \\ 0 & \cos(\alpha_1+\alpha_2+\alpha_3) & -\sin(\alpha_2+\alpha_3+\alpha_5) & d_1-d_3\sin(\alpha_2+\alpha_3)-d_4\sin(\alpha_2+\alpha_3)-d_2\sin(\alpha_2)-d_5\sin(\alpha_2+\alpha_3+\alpha_5) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matlab code of forward kinematics:

```
%% Forward kinematics
clear all;
clc;

syms th1;
syms d1 d2 d3 d4 d5;
syms alpha2 alpha3 alpha5;

alpha1 = 90;

%% A1
% Rot z, theta
A1_1 = [cos(th1) -sin(th1) 0 0;
        sin(th1)  cos(th1) 0 0;
        0         0        1 0;
        0         0        0 1];

% Trans z, d1
A1_2 = [1 0 0 0;
        0 1 0 0;
        0 0 1 d1;
        0 0 0 1];

% Rot x, alpha1
A1_3 = [1 0 0 0;
```

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```
        0 cos(alpha1) -sin(alpha1) 0;
        0 sin(alpha1)  cos(alpha1) 0;
        0 0 0 1];

A1_4 = [1 0 0 0;
        0 0 -1 0;
        0 1  0 0;
        0 0  0 1];

A1 = A1_1 * A1_2 * A1_4;

%% A2
% Rot x, alpha2
A2 = [1 0 0 0;
      0 cos(alpha2) -sin(alpha2) 0;
      0 sin(alpha2)  cos(alpha2) 0;
      0 0 0 1];

%% A3
% Trans z, d2
A3_1 = [1 0 0 0;
        0 1 0 0;
        0 0 1 d2;
        0 0 0 1];

% Rot x, alpha3
A3_2 = [1 0 0 0;
        0 cos(alpha3) -sin(alpha3) 0;
        0 sin(alpha3)  cos(alpha3) 0;
        0 0 0 1];

A3 = A3_1 * A3_2;

%% A4
% Trans z, d3
A4 = [1 0 0 0;
      0 1 0 0;
      0 0 1 d3;
      0 0 0 1];

%% A5
% Trans z, d4
A5_1 = [1 0 0 0;
        0 1 0 0;
        0 0 1 d4;
        0 0 0 1];

% Rot x, alpha5
A5_2 = [1 0 0 0;
        0 cos(alpha5) -sin(alpha5) 0;
```

```
        0 sin(alpha5)  cos(alpha5) 0;
        0 0 0 1];

A5 = A5_1 * A5_2;

%% A6
% Trans z, d5
A6 = [1 0 0 0;
      0 1 0 0;
      0 0 1 d5;
      0 0 0 1];

%% Global transform
T = A1 * A2 * A3 * A4 * A5 * A6;
T = simplify(T);
T
```

b) Inverse

Firstly, I had to create specific equations from the final matrix.

$$x = \sin(\theta_1)(d_3\cos(\alpha_2+\alpha_3) + d_4\cos(\alpha_2+\alpha_3) + d_2\cos(\alpha_2) + d_5\cos(\alpha_2+\alpha_3+\alpha_5))$$

$$y = -\cos(\theta_1)(d_3\cos(\alpha_2+\alpha_3) + d_4\cos(\alpha_2+\alpha_3) + d_2\cos(\alpha_2) + d_5\cos(\alpha_2+\alpha_3+\alpha_5))$$

$$z = d_1 - d_3\sin(\alpha_2+\alpha_3) - d_4\sin(\alpha_2+\alpha_3) - d_2\sin(\alpha_2) - d_5\sin(\alpha_2+\alpha_3+\alpha_5)$$

Then once again I used MatLab software:

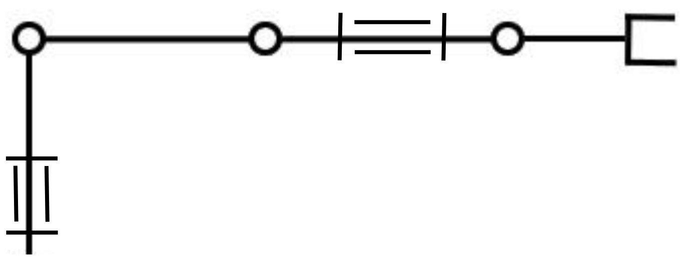
```
%% Inverse kinematics
clc

x0=0;
y0=0;
z0=0;
% Setting needed data
syms alpha2 alpha3 alpha5
th1 = 0;
d1 = 1000;
d2 = 1500;
d3 = 1200;
d4 = 0;
d5 = 500;

% Position equations
x = sin(th1)*(d3*cos(alpha2 + alpha3) + d4*cos(alpha2 + alpha3) + d2*cos(alpha2) + d5*cos(alpha2 +
alpha3 + alpha5)) == x0;
```

```
y = -cos(th1)*(d3*cos(alpha2 + alpha3) + d4*cos(alpha2 + alpha3) + d2*cos(alpha2) + d5*cos(alpha2
+ alpha3 + alpha5)) == y0;
z = d1 - d3*sin(alpha2 + alpha3) - d4*sin(alpha2 + alpha3) - d2*sin(alpha2) - d5*sin(alpha2 + alpha3 +
alpha5) == z0;

% ^ x0, y0, z0 come from forward kinematics

% Solving position equations
result = solve([x, y, z], [alpha2, alpha3, alpha5]);

a2_rad = double(result.alpha2);
a3_rad = double(result.alpha3);
a5_rad = double(result.alpha5);

% Converting radians to degrees
a2_deg = a2_rad * (180/pi);
a3_deg = a3_rad * (180/pi);
a5_deg = a5_rad * (180/pi);
```

V. DIFFERENT CONFIGURATIONS

a) Configuration I

The configuration on the scheme:



- *FORWARD KINEMATICS*

Data assumptions for the configuration:

$d1$=0 $\theta 1$=0°

$d2$= 560 $\alpha 2$=0°

$d3$= 440 $\alpha 3$=0°

$d4$= 390 $\alpha 5$=0°

$d5$= 520

I used the MatLab extension to obtain faster transverse matrix:

```
T =

     1     0     0     0
     0     0    -1 -3200
     0     1     0  1000
     0     0     0     1
```

- *INVERSE KINEMATICS*

Also obtained the values in degrees using MatLab:

```
     alpha_2          alpha_3          alpha_5
a2_deg =              a3_deg =                  a5_deg =

   1.0e+02 *

  -0.2274 + 0.0000i   0.4045 + 0.1885i   0.0000 - 0.6099i
   0.0000 - 0.4521i   0.0000 + 0.9657i  -1.8000 - 0.6099i
   0.0000 + 1.3420i   1.8000 + 0.0000i  -1.8000 - 2.4004i
   1.8000 + 0.0000i  -1.8000 - 0.7752i   0.0000 + 2.0200i
   0.0000 - 0.2836i  -0.2262 + 0.6069i   0.9000 + 0.0000i
   0.0000 + 0.2836i  -0.2262 - 0.6069i   0.9000 + 0.0000i
   0.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i
   0.0000 + 0.3971i   0.0000 + 1.1149i   1.8000 + 0.0000i
   0.0000 - 0.3971i   0.0000 - 1.1149i   1.8000 + 0.0000i
   0.2274 + 0.0000i  -0.4045 - 0.1885i   0.0000 + 0.6099i
   0.0000 - 0.4341i   0.0000 + 0.9632i  -1.8000 - 0.5016i
   0.0000 + 0.4341i   0.0000 - 0.9632i   1.8000 + 0.5016i
   0.0000 + 0.4521i   0.0000 - 0.9657i   1.8000 + 0.6099i
   0.0000 - 1.3420i   1.8000 + 0.0000i   1.8000 + 2.4004i
   1.8000 + 0.0000i   1.8000 + 0.7752i   0.0000 - 2.0200i
```

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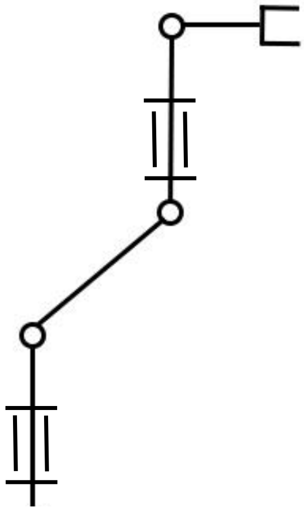
b) Configuration II



- *FORWARD KINEMATICS*

Data:

$d1$= 0 $\theta 1$=0°

$d2$=560 $\alpha 2$=−30°

$d3$=440 $\alpha 3$=−60°

$d4$=390 $\alpha 5$=90°

$d5$=520

Transverse matrix:

```
T =

     1     0     0     0
     0     0    -1  -1799
     0     1     0   2950
     0     0     0     1
```

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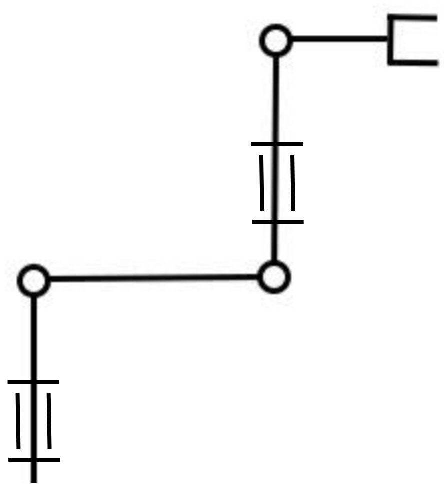
- *INVERSE KINEMATICS*

```
      alpha_2           alpha_3           alpha_5
a2_deg =                  a3_deg =                   a5_deg =

   1.0e+02 *

  -0.4731 + 1.2279i   1.8000 + 0.0000i  -1.8000 - 2.1764i
  -1.7249 - 0.4043i   1.3595 + 0.9459i   0.2049 - 1.8150i
   0.7787 + 0.4043i  -1.5494 + 0.4103i   0.2049 - 1.8150i
   1.8000 + 0.0000i   1.4941 - 0.6543i   0.0000 + 1.7609i
  -0.4731 + 0.2609i   0.0000 - 0.7699i   1.8000 + 0.0000i
  -0.4731 - 1.2279i   1.8000 + 0.0000i   1.8000 + 2.1764i
   0.0000 + 0.0000i  -1.4941 - 0.6543i   0.0000 + 1.7609i
   1.8000 + 0.0000i   1.4941 + 0.6543i   0.0000 - 1.7609i
  -0.4731 - 0.3267i   0.0000 + 0.6699i  -1.8000 - 0.5016i
  -0.4731 + 0.3267i   0.0000 - 0.6699i   1.8000 + 0.5016i
   0.0000 + 0.0000i  -1.4941 + 0.6543i   0.0000 - 1.7609i
  -0.4731 - 0.2609i   0.0000 + 0.7699i   1.8000 + 0.0000i
  -0.1082 + 0.0000i  -0.6814 + 0.0000i   0.0000 + 0.0000i
  -0.8380 + 0.0000i   0.6814 + 0.0000i   0.0000 + 0.0000i
```

c) <u>Configuration III</u>



Data:

$d1$=0 $\theta 1$=0°

$d2$=560 $\alpha 2$=0°

$d3$=440 $\alpha 3$=90°

$d4$=390 $\alpha 5$=−90°

$d5$=520

Transverse matrix:

```
T =

     1     0     0     0
     0     0    -1 -2000
     0     1     0  2200
     0     0     0     1
```

- *INVERSE KINEMATICS*

```
      alpha_2            alpha_3            alpha_5
a2_deg =                    a3_deg =                    a5_deg =

   1.0e+02 *

  -0.3096 - 1.1476i   1.8000 + 0.0000i   1.8000 + 2.0200i
   1.8000 + 0.0000i   1.6108 - 0.6333i   0.0000 + 1.7150i
   1.8000 + 0.0000i   1.6108 + 0.6333i   0.0000 - 1.7150i
  -0.7268 + 0.0000i   0.6712 + 0.0000i   0.4967 + 0.0000i
   0.1075 + 0.0000i  -0.9521 + 0.0000i   0.4967 + 0.0000i
  -0.3096 + 1.1476i   1.8000 + 0.0000i  -1.8000 - 2.0200i
  -0.7765 + 0.0000i   0.8663 + 0.0000i   0.0000 + 0.0000i
   0.1572 + 0.0000i  -0.8663 + 0.0000i   0.0000 + 0.0000i
  -0.3096 - 0.2529i   0.0000 + 0.4371i  -1.8000 - 0.5016i
  -0.3096 + 0.2529i   0.0000 - 0.4371i   1.8000 + 0.5016i
  -0.3096 - 0.1376i   0.0000 + 0.4234i   1.8000 + 0.0000i
  -0.3096 + 0.1376i   0.0000 - 0.4234i   1.8000 + 0.0000i
```

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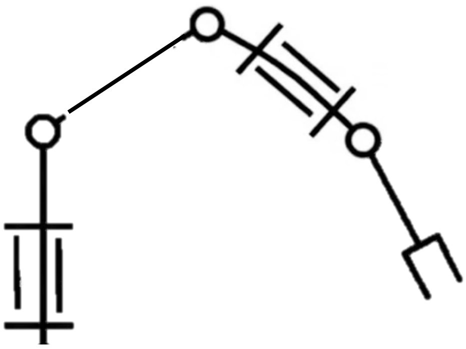
d) <u>Configuration IV</u>



Data:

$d1$=0 $\theta 1$=0°

$d2$=560 $\alpha 2$=45°

$d3$=440 $\alpha 3$=30°

$d4$=390 $\alpha 5$=-45°

$d5$=520

Transverse Matrix

```
T =

    1.0000         0         0  509.8602
         0    0.7071   -0.7071   22.3045
         0    0.7071    0.7071  395.9798
         0         0         0    1.0000
```

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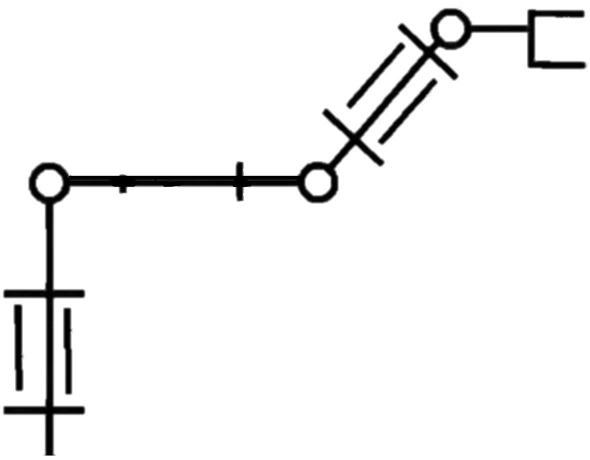
- *INVERSE KINEMATICS*

```
      alpha_2          alpha_3          alpha_5
a2_deg =                 a3_deg =                 a5_deg =

   1.0e+02 *

   1.2419 + 1.5904i  -4.5975 + 0.0000i  -0.7240 + 1.1940i
   1.2419 - 1.5904i   0.8298 + 0.7441i  -0.7240 - 1.1940i
  -0.1676 + 1.4111i   0.8298 - 0.7441i   0.6589 + 1.1791i
  -0.1676 - 1.4111i  -0.3595 + 0.9441i   0.6589 - 1.1791i
  -0.0743 + 0.2351i  -0.3595 - 0.9441i   0.7606 + 0.0000i
  -0.0743 - 0.2351i  -0.3432 + 0.0000i   0.3696 + 0.0000i
```

e) Configuration V



Data:

$d1$=0 $\theta1$=0°

$d2$=560 $\alpha2$=-90°

$d3$=440 $\alpha3$=10°

$d4$=390 $\alpha5$=120°

$d5$=520

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Transverse matrix:

```
T =

    0.9848         0    0.1736  993.3154
   -0.1736         0    0.9848 -390.0000
         0   -1.0000         0 -110.0000
         0         0         0    1.0000
```

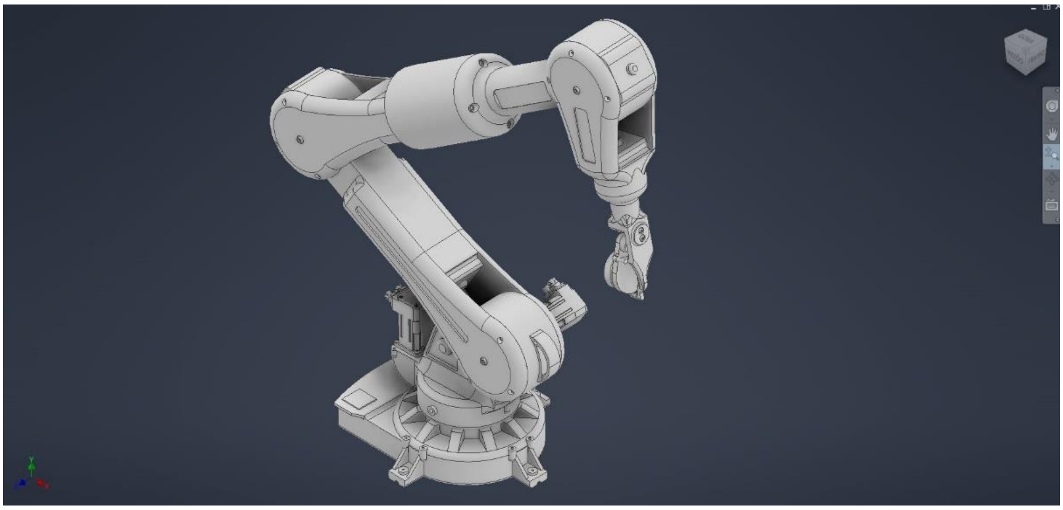
- *INVERSE KINEMATICS*

```
      alpha_2          alpha_3          alpha_5
a2_deg =                 a3_deg =                 a5_deg =

   1.0e+02 *

  -2.4720 + 0.0000i   0.8893 + 0.6784i  -0.7390 + 0.2799i
  -0.1330 + 0.6338i   0.8893 - 0.6784i  -0.7390 - 0.2799i
  -0.1330 - 0.6338i  -0.1350 + 0.6169i  -0.0504 + 0.6060i
   0.1817 + 0.2955i  -0.1350 - 0.6169i  -0.0504 - 0.6060i
   0.1817 - 0.2955i   0.3914 + 0.0000i   0.2894 + 0.4322i
  -0.1253 + 0.0000i  -0.4001 + 0.0000i   0.2894 - 0.4322i
```

VI. 3D MODEL CREATED IN AUTO DESK INVENTOR



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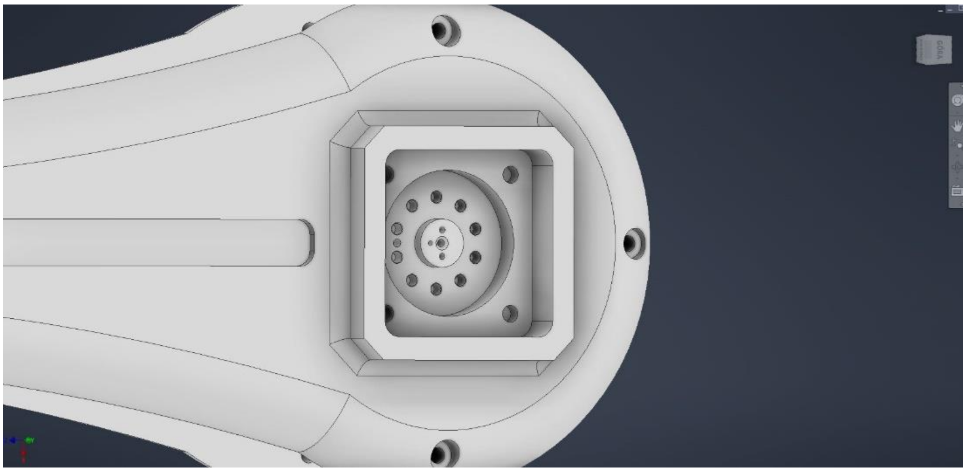
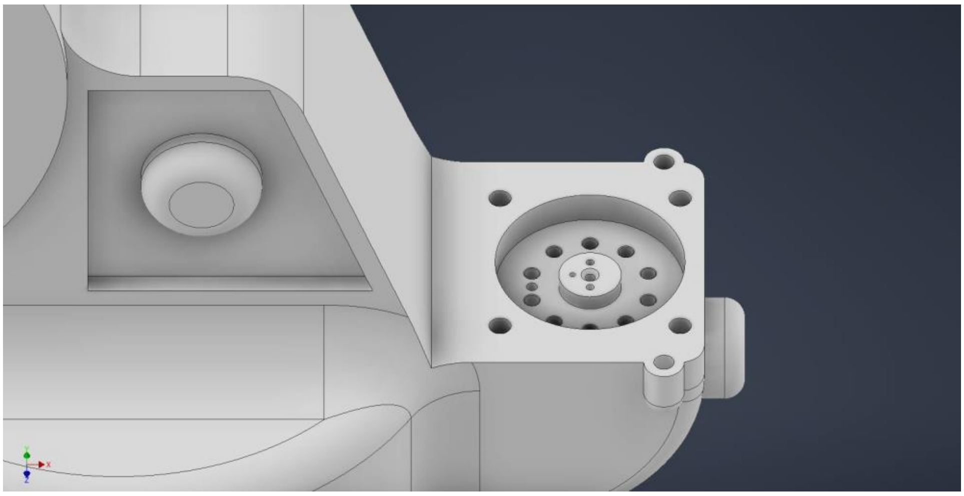
Disclaimer: Screenshots of DriveSpins connections with robot's modules.





Mounting manipulator of Lidar Sensor

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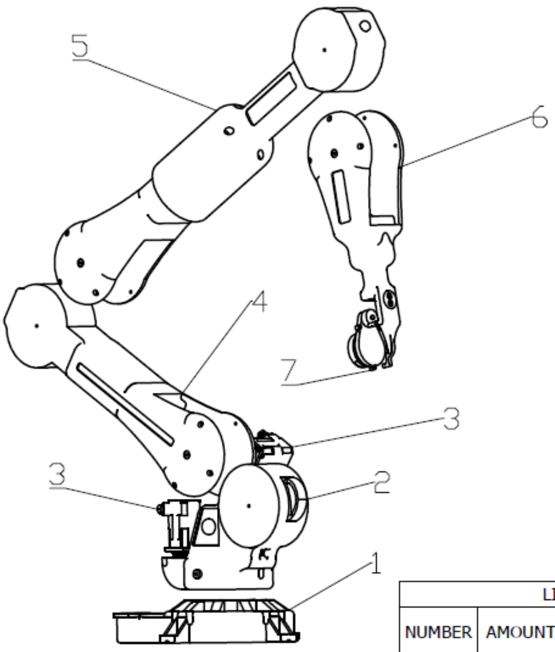
All models are in the ZIP FILE

VII. ASSEMBLY DRAWING OF A WHOLE MODEL



LIST OF PARTS

NUMBER	AMOUNT	NAME OF A PART	DETAILS
1	1	module1_1	
2	1	module2_2	
3	2	ds_050_-_63_-_1	
4	1	module3	
5	1	module4	
6	1	module5	
7	1	LIDAR X4 PRO	

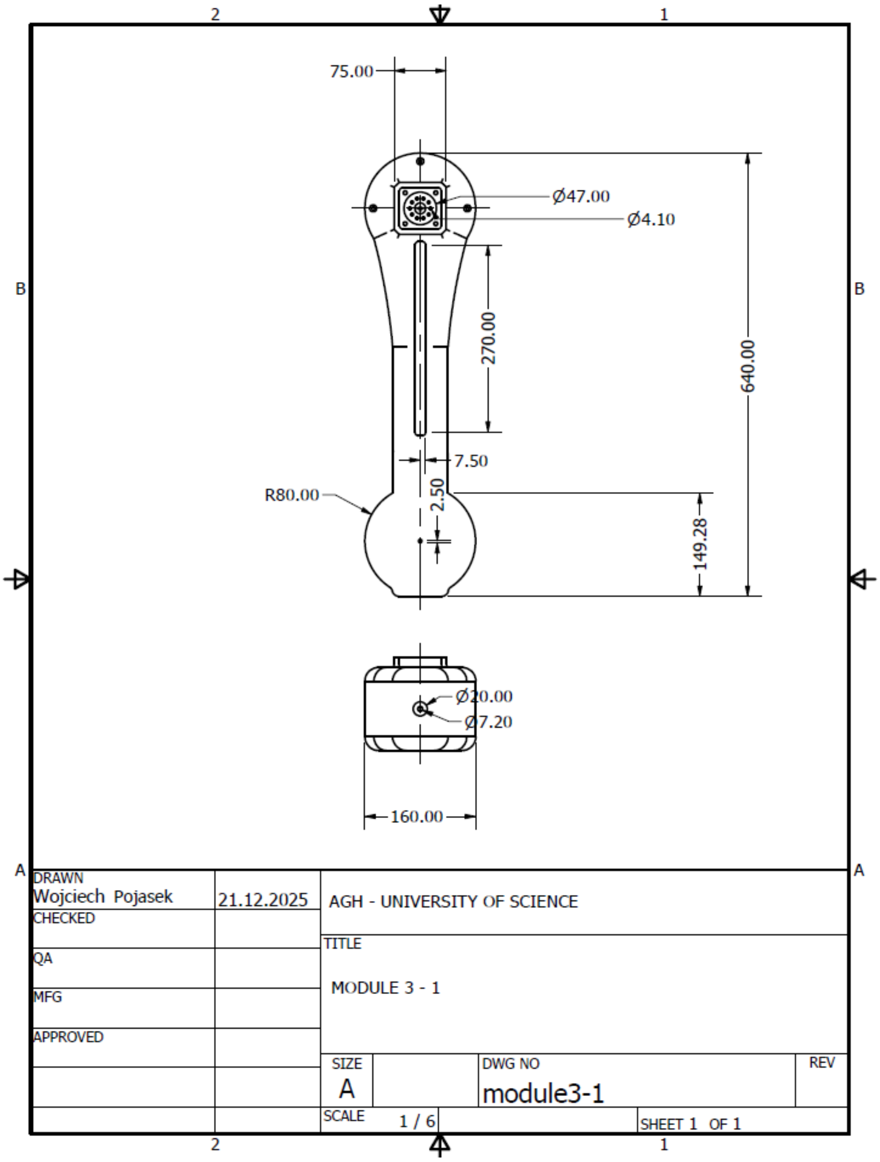
DRAWN Wojciech Pojasek	21.12.2025	
CHECKED		TITLE ASSEMBLY DRAWING OF A ROBOT - SW_POJASEK
QA		
MFG		
APPROVED		
		SIZE A	DWG NO assembly_final	REV
		SCALE 1 / 8	SHEET 1 OF 2

Pojasek Wojciech

Index Number: 419773

VIII. 2D DRAWINGS OF SPECIFIC PARTS

First Part – Module III



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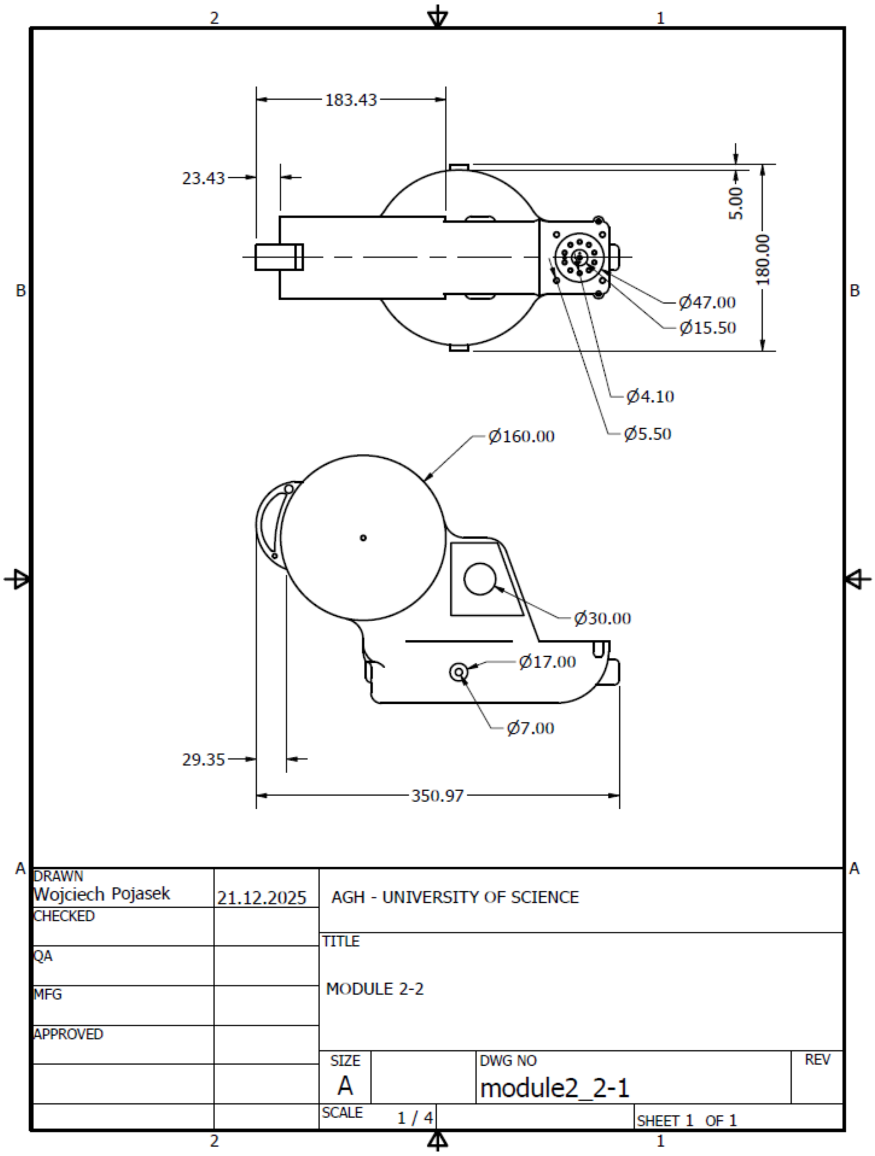
<u>Second Part – Module II</u>



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Index Number: 419773